

Bilingual Textbook Question-Answering System

Goal: Build an AI assistant where a student asks a question and the system answers first from the textbook itself, then gives a simple explanation and one or more examples. The system must support **both Arabic and English books**, because some student questions come from Arabic textbooks and others come from English textbooks.

Prepared on April 15, 2026. This report covers bilingual Arabic-English textbook QA, retrieval, explanation generation, model choices, public datasets, and fine-tuning.

1. Does This Require AI?

Yes. The full product is a combination of normal software and AI.

- **Normal software** handles upload, storage, database, user accounts, interface, language routing, and logging.
- **AI** handles semantic retrieval, question understanding, answer grounding, explanation generation, and example generation.

Best framing: This is not training a model from zero. It is usually a **Retrieval-Augmented Generation (RAG)** system built on top of pretrained Arabic-capable and English-capable models.

2. Bilingual Requirement: Arabic and English

Your final system should support two cases:

1. **Arabic question + Arabic textbook**
2. **English question + English textbook**

A stronger version can also support:

1. **Arabic question + English textbook**
2. **English question + Arabic textbook**

Recommended first scope: start with same-language retrieval first. That means Arabic questions search Arabic books, and English questions search English books. Cross-language retrieval can be added later if needed.

Recommended bilingual output policy

- If the source book is Arabic, the quoted answer should stay in Arabic.
- If the source book is English, the quoted answer should stay in English.
- The explanation can follow the student's preferred language.
- The final UI can include a language selector: `Arabic` , `English` , or `same as source` .

3. Recommended System Design

For your use case, the safest design is:

1. **Detect language** of the question or use the chosen book language.
2. **Retrieve** the relevant paragraph or section from the correct textbook collection.
3. **Extract** the exact answer span or best supporting sentence from that passage.
4. **Generate** a simple explanation and example using only that evidence.

Why this order matters: If the model explains first without grounding itself in the book, it may hallucinate. For educational use, a quote-first, explain-second pipeline is much safer.

Output format

1. **From the book:** exact quote or faithful extract
2. **Explanation:** simpler explanation in Arabic or English
3. **Example:** one short example
4. **Source:** chapter, section, page

4. Technical Architecture

Module	Role
Frontend	Student submits Arabic or English questions and sees structured answers with source citations.
Language router	Detects Arabic or English and routes the request to the correct retrieval pipeline.
Book ingestion	Extracts text from Arabic or English PDF/EPUB, cleans it, chunks it, and stores metadata.
Embedding service	Converts each chunk into a semantic vector using multilingual or language-specific embeddings.
Retriever	Finds the most relevant chunks using embeddings, keywords, or hybrid search.
Reader / answer module	Selects exact supporting text and generates explanation plus example.
Verifier	Checks whether the answer is truly supported by retrieved text.
Databases	Stores users, books, chunks, logs, and vector embeddings.

Flow

Arabic/English textbook -> text extraction -> chunking -> embeddings -> vector database

Arabic/English question -> language routing -> retrieval -> evidence selection -> answer generation -> grounded response

5. Language-Specific Challenges

- **Arabic normalization:** diacritics, alef variants, ta marbuta, ya/alef maqsura, punctuation.
- **Arabic PDF extraction quality:** Arabic PDFs can have line-order and shaping issues.
- **English textbook structure:** tables, captions, formulas, and side-notes may affect chunking.
- **Mixed-language education:** a student may ask in Arabic about an English book or vice versa.
- **Terminology mismatch:** scientific terms may appear in English inside Arabic questions.

6. Best Build Strategy

Phase 1: Build a bilingual RAG system with pretrained models. Do not train a base model from scratch.

Phase 2: If needed, fine-tune either:

- a bilingual retriever for textbook search,
- an extractive QA model for exact answer spans,
- or an instruction/generation model for better explanations.

7. Which Models Should You Use?

Yes, the models are pretrained. In modern NLP, you almost always start from pretrained models and then adapt them.

7.1 Recommended model stack for the first version

Task	Recommended type	Examples	Pretrained?
Embeddings / retrieval	Multilingual sentence embedding model	multilingual-e5 , bge-m3 , multilingual sentence-transformer variants	Yes
Arabic exact answer extraction	Arabic encoder fine-tuned for extractive QA	AraBERT , ARBERT , MARBERT , CAMELBERT	Yes
English exact answer extraction	English encoder fine-tuned for extractive QA	BERT , RoBERTa , DeBERTa	Yes
Explanation generation	Multilingual or bilingual generative model	mT5 , AraT5 , multilingual LLM/API model	Yes
Optional reranker	Cross-encoder or reranking model	Arabic, English, or multilingual cross-encoder	Yes

7.2 Practical recommendation

- **Fastest MVP:** one multilingual embedding model + one multilingual generative model.
- **More accurate academic version:** multilingual retriever + Arabic QA model + English QA model + generator.

This means your system can first find the passage, then extract the answer, then explain it.

8. Can You Train a New Model?

Yes, but there are three different meanings of train:

1. **Use pretrained model only:** no new training, only prompting and retrieval.
2. **Fine-tune a pretrained model:** most realistic for a student project.
3. **Pretrain a brand-new model from scratch:** usually too expensive and unnecessary.

For your project, **fine-tuning pretrained models** is the right path if the basic RAG system is not enough.

9. What Data Do You Need?

9.1 Minimum data for an MVP

- Arabic and English textbook files
- metadata: title, chapter, section, page, language
- a set of sample student questions in Arabic and English

9.2 Data for fine-tuning

Training target	Needed data format
Retriever	(question, positive passage, negative passages)
Extractive QA model	(context, question, answer_text, start_char)
Generator / explainer	(question, evidence passage, target answer with explanation)
Instruction tuning	(instruction, input, output)

9.3 If you cannot collect data from your own books

That is completely workable. You can start with **public internet datasets** for Arabic and English, then later test on your book collection through RAG.

- **Arabic QA datasets:** ARCD, Arabic-SQuAD, TyDi QA Arabic, MLQA Arabic, XQuAD Arabic.
- **English QA datasets:** SQuAD, Natural Questions, HotpotQA, RACE, NarrativeQA.
- **Multilingual QA datasets:** TyDi QA, MLQA, XQuAD, XOR QA.
- **Retrieval datasets:** MS MARCO for English warm-start, multilingual retrieval benchmarks, BEIR-style tasks where available.

Important limitation: public internet datasets are useful for fine-tuning general question answering, but they are not the same as your exact textbook domain. So the best plan is to use public data for warm-start training and rely on RAG over your actual books during inference.

9.4 Suggested online data sources

Dataset	Language	Use	Where to get it
ARCD	Arabic	Extractive QA	Hugging Face or ACL paper resources
Arabic-SQuAD	Arabic	Extractive QA	Often distributed through research repos and dataset hubs
TyDi QA	Multilingual including Arabic and English	Multilingual QA	Google Research / Hugging Face
MLQA	Multilingual including Arabic and English	Cross-lingual QA	Hugging Face / original repo
XQuAD	Multilingual including Arabic and English	Evaluation and warm-start	Hugging Face / original repo
SQuAD	English	Extractive QA	Stanford / Hugging Face
Natural Questions	English	Open-domain QA	Google Research
RACE	English	Reading comprehension	Dataset hubs and original paper resources

9.5 Example row for a custom bilingual training sample

```
{
  "language": "ar",
  "book_id": "physics_grade10",
  "chapter": "?????",
  "page": 42,
  "question": "?? ?? ????????",
  "evidence": "?????? ?? ??? ???? ????? ?????? ?????.",
  "quoted_answer": "?????? ?? ??? ???? ?????? ?????? ?????.",
  "explanation": "???? ?? ?? ??? ???? ???? ?? ??? ?? ??? ?????.",
  "example": "??? ??? ???? ?????? ?? 20 ?? 40 ??? ?????? ??? ???? ???? ?????."
}

{
  "language": "en",
  "book_id": "biology_unit3",
  "chapter": "Cells",
  "page": 18,
  "question": "What is diffusion?",
  "evidence": "Diffusion is the movement of particles from an area of higher concentration to an area of lower concentration.",
  "quoted_answer": "Diffusion is the movement of particles from an area of higher concentration to an area of lower concentration.",
  "explanation": "This means particles naturally spread out from crowded places to less crowded places.",
  "example": "Perfume smell spreading across a room is an example of diffusion."
}
```

10. Fine-Tuning Plan

10.1 Fine-tuning the retriever

Goal: retrieve the correct Arabic or English textbook paragraph.

1. Build question-passage pairs from public QA datasets and any accessible educational text.
2. For each question, mark one correct paragraph and several hard negatives.
3. Train a bi-encoder using contrastive loss.
4. Evaluate with Recall@k and MRR.

10.2 Fine-tuning the extractive QA model

Goal: return the literal answer from the source passage.

1. Format Arabic and English context-question pairs.
2. Mark exact answer spans inside the context.
3. Fine-tune Arabic and English span-extraction models, or use one multilingual model.
4. Evaluate with Exact Match and F1.

10.3 Fine-tuning the generator

Goal: produce a simple explanation and an example in Arabic or English.

1. Create supervised examples where input includes the question and evidence passage.
2. Output should follow your final template: quote, explanation, example, source.
3. Fine-tune a sequence-to-sequence model such as AraT5 or mT5, or use parameter-efficient tuning on a multilingual LLM.
4. Evaluate for grounding, clarity, and hallucination rate.

11. Step-by-Step Fine-Tuning Workflow

1. **Collect data:** gather public Arabic and English QA datasets from the internet.
2. **Clean and normalize:** Arabic normalization plus English cleanup and consistent punctuation.
3. **Prepare retrieval pairs:** generate positives and hard negatives.
4. **Prepare QA labels:** exact answer spans for Arabic and English.
5. **Prepare generation labels:** target answer with explanation and example.
6. **Split the dataset:** train, validation, test with no leakage.
7. **Choose the training target:** retriever, reader, generator, or all three.
8. **Start with the retriever:** retrieval quality usually determines overall system quality.
9. **Then fine-tune extractive QA** for exact textbook-style answers.
10. **Then adapt the generator** for teaching-style explanations.
11. **Run grounded evaluation:** check citation correctness and unsupported claims.
12. **Deploy and log:** save failed questions to improve the next version.

12. How Much Data Is Enough?

Component	Small useful start	Better target
Retriever	1,000 to 3,000 question-passage pairs	5,000+
Extractive QA	1,000 to 2,000 labeled spans	5,000+
Explanation generator	500 to 1,500 high-quality supervised examples	3,000+

If your data is smaller than this, do not worry. A strong RAG pipeline can still work well, especially if your textbook collection is limited and domain-specific.

13. How Long Does Training Take?

The answer depends on model size, dataset size, hardware, and whether you are fine-tuning the full model or using LoRA / parameter-efficient tuning.

Training target	Typical setup	Estimated time
Embedding retriever fine-tuning	Small to medium bi-encoder on 1 GPU	2 to 8 hours
Arabic or English extractive QA fine-tuning	BERT-size model on 1 GPU	1 to 6 hours
Multilingual extractive QA fine-tuning	mBERT / XLM-R size on 1 GPU	4 to 12 hours

Small seq2seq explanation model	mT5-small/base or AraT5-like model on 1 GPU	4 to 16 hours
LoRA fine-tuning of a larger LLM	7B-class model on strong GPU	6 to 24 hours
Training from scratch	Large-scale pretraining	weeks to months

Realistic recommendation for a graduation project: plan for days, not months, by using pretrained models and fine-tuning only the necessary components. Full pretraining from scratch is not realistic unless you have major compute resources.

Practical estimate: if you fine-tune a retriever, an Arabic QA model, and an English QA model on a normal research GPU, the full experimental cycle may take roughly **2 to 7 days** including preprocessing, training, validation, reruns, and debugging.

14. Recommended Software Stack

- **Frontend:** React or Next.js
- **Backend:** FastAPI
- **Vector store:** Chroma or FAISS
- **Relational database:** PostgreSQL
- **PDF extraction:** PyMuPDF or pdfplumber
- **Training/inference:** PyTorch + Hugging Face Transformers
- **Evaluation:** custom scripts for retrieval accuracy, EM/F1, and grounding checks

15. Research Papers and PDF Links

The papers below are the most relevant starting set for your topic.

Paper	Why it matters	PDF / source link
Lewis et al., 2020, <i>Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks</i>	Foundational RAG paper.	PDF
Karpukhin et al., 2020, <i>Dense Passage Retrieval for Open-Domain Question Answering</i>	Strong retriever design for QA.	PDF
Kim et al., 2019, <i>Textbook Question Answering with Multi-modal Context Graph Understanding and Self-supervised Open-set Comprehension</i>	Directly about textbook QA.	PDF
Alawwad et al., 2024, <i>Enhancing Textbook Question Answering Task with Large Language Models and Retrieval Augmented Generation</i>	Very close to your project idea.	PDF
Slobodkin et al., 2024, <i>Attribute First, then Generate: Locally-attributable Grounded Text Generation</i>	Supports your quote-first, explain-second design.	PDF
Mozannar et al., 2019, <i>Neural Arabic Question Answering</i>	Arabic open-domain QA and ARCD dataset.	PDF
Antoun et al., 2020, <i>AraBERT: Transformer-based Model for Arabic Language Understanding</i>	Key Arabic encoder model.	PDF
Abdul-Mageed et al., 2021, <i>ARBERT & MARBERT: Deep Bidirectional Transformers for Arabic</i>	Strong Arabic pretrained encoders.	PDF
Nagoudi et al., 2022, <i>AraT5: Text-to-Text Transformers for Arabic Language Generation</i>	Arabic generation and seq2seq fine-tuning.	PDF
Ghaddar et al., 2022, <i>Revisiting Pre-trained Language Models and their Evaluation for Arabic NLP</i>	Good overview of Arabic PLMs and evaluation.	PDF
Rajpurkar et al., 2016, <i>SQuAD: 100,000+ Questions for Machine Comprehension of Text</i>	Foundational English extractive QA dataset.	PDF
Clark et al., 2020, <i>TyDi QA: A Benchmark for Information-Seeking Question Answering in Typologically Diverse Languages</i>	Useful multilingual benchmark including Arabic and English.	PDF

16. Best Final Recommendation for Your Project

1. Start with a **bilingual RAG pipeline** using pretrained models.
2. Use **multilingual embeddings** so Arabic and English are both supported.
3. Keep the output strictly structured: quote, explanation, example, source.
4. Use public internet datasets first, because you cannot annotate your own books right now.
5. Fine-tune the **retriever first**, because retrieval quality is critical.
6. Fine-tune Arabic and English extractive QA models if you need exact source spans.
7. Fine-tune the generator last for educational explanation style.

17. Suggested Project Roadmap

1. Collect Arabic and English textbooks.
2. Extract and clean text.
3. Build chunking plus metadata pipeline with a language field.
4. Store multilingual embeddings in Chroma or FAISS.
5. Implement language routing and grounded answering.
6. Download public Arabic and English QA datasets from the internet.
7. Prepare a bilingual evaluation set.
8. Measure retrieval quality and answer groundedness.
9. Fine-tune components if needed.

18. Bottom Line

Yes, you can build this. For Arabic and English textbooks, the strongest practical route is:

Bilingual book ingestion + multilingual retrieval + exact quoted answer + explanation + citations.

That means you are building an **AI system on top of normal software**, not only a search engine and not a model trained from zero.